Tac & Lock — Field Manual

Version 1.6.1 · 2300AD Space Combat Simulator

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1. Overview

Tac & Lock is a browser-based Virtual Tabletop (VTT) tool for running **2300AD** space combat at the gaming table. It is GM-operated and designed for shared-screen play — one person drives, everyone watches.

Rules source: **2300AD Core Book 3: Vehicles and Spacecraft** pp.52–62 (primary). **Traveller 2022 Core Rulebook** is used only for internal critical hit tables (Location table p.169, Effects table p.170 in this project's specific CRB PDF — B3 p.58 itself cites

"p.158–159", verified wrong by direct extraction, issue #23/#67) and weapon traits (p.79). Where B3 and CRB diverge, **B3 wins**.

Combat uses **range bands** — no hex grid, no velocity vectors. The seven bands (Adjacent → Distant) define all spatial relationships between ships. Ships move between bands via an opposed Pilot check (Open/Close); TAC Speed is a fixed DM added to the roll, never spent.

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2. Dashboard

The Dashboard is the pre-battle lobby. Left panel: ship profiles. Right panel: operations console + tactical display.

2.1 Ship Profiles

Control	Action
+ NEW PROFILE	Create a new ship from scratch.
✎ Edit	Modify an existing profile.
⧉ Duplicate	Clone a profile as a starting point.
⊗ Delete	Remove a profile (confirmation required).
↓ IMPORT	Load profiles from a `.json` file.
↑ EXPORT	Save all profiles to a `.json` file.
📖 OFFICIAL CATALOG	Browse canonical 2300AD vessels and add them to profiles.

2.2 Tactical Display

The right panel mirrors the current session at a glance, even before entering battle.

Field	Notes
PROTOCOL / RANGE SYSTEM	Fixed ruleset identifiers.
ROUND / PHASE	Current round and phase of the active session.
VESSELS / DRONES	Ship and drone/missile counts in the active roster.
SESSION / SAVED	Session name and timestamp of the last autosave.
SHIP ROSTER	Ships grouped by faction (Players / NPC / Neutral). Each row shows the ship's token silhouette, name, class, TAC Speed / Armour, range to the nearest enemy, a status badge (DOGFIGHT when Adjacent/Close, DESTROYED, or COMBAT/NEUTRAL), and hull bar.

2.3 Operations Console

Control	Action
▶ ENTER BATTLE	Enter the battle screen (keeps existing roster).
↓ RESUME FROM FILE	Load a previously saved `.json` session. A preview shows the roster before you confirm.
💾 SAVE	Download the current session as `.json`.
CLEAR	Reset battle state (confirmation required).
📖 FIELD MANUAL	Opens this manual.

2.4 Ship Profile Form

Each profile stores the ship's combat statistics, weapons, sensors, computer, software, and crew.

Field	Notes
Name / Class	Display name and vessel class.
Hull Points	Maximum hull points.
Armour	Reduces incoming damage.
TAC Speed	Tactical Speed rating — how many band-change points the ship generates per round.
Signature	Base electromagnetic signature. Modified dynamically in battle.
Sensors	Type (Basic Military / Improved / Advanced) and DM bonus.
Computer	Model and bandwidth.
Weapons	Add weapon slots — choose from the canonical 2300AD weapon list.
Software	Select installed software (Fire Control, Auto-Repair, etc.).
Crew	Named crew members with skill ratings (Tactics, Pilot, Engineer, Gunner, Electronics).

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3. Round & Turn Structure

There is no "Manoeuvre Step / Attack Step / Actions Step" cycle in 2300AD — that's a Traveller CRB structure 2300AD B3 doesn't use. There are only three **stages**, shown in the HUD:

Stage	What happens
Setup	Add ships to the battle (right-click background → Add ship).
Initiative	Roll opposed Tactics(naval) check — order fixed for the entire engagement.
Combat	Ships act in initiative order, one ship's turn at a time, round after round.

During **Combat**, a ship's turn is open-ended. While a ship is the current actor, the GM can freely open **Manoeuvre…**, **Attack…**, **Launch Drone…**, or **Crew Action…** for it, in any order, as many times as that ship's crew have budget for. Each crew **role** — not the ship as a whole — has its own action budget for the round, equal to that crew member's skill level in the role's primary skill (Pilot, Gunner, Engineer, etc.). Gunnery is always capped at 1 action per round regardless of skill (Fire Weapon / Deploy-Recharge Screens / Point Defence / Operate UTES Array share that single use); Pilot and most other roles can act multiple times.

The HUD shows the current actor and each role's remaining budget. When the GM is done with that ship's turn — with or without spending every remaining action — click **END SHIP'S TURN ⟶** to pass to the next ship in initiative order. Once every ship has had its turn, click **NEXT ROUND ⟶** to reset action budgets and start the next round.

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4. Setup Phase

Right-click the background → **Add ship**. A modal opens:

Field	Notes
Profile	Which saved ship profile to use.
Faction	Players / NPC / Neutral — determines card colour coding.
Token Shape	Frigate / Scout / Freighter / Courier / Alien — hull silhouette shown on the bento card, pre-selected from the profile's catalog category.

Field	Notes
Initial Range Band	The starting distance from all other ships (usually Long).

Ships are shown as bento cards grouped by faction, each with a ship-token silhouette. Each card shows hull bar, TAC Speed, armour, effective signature, weapons, critical tracks, and inbound drone ETA.

Right-click a card to open the context menu. Available actions depend on phase and initiative turn.

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5. Initiative

Formula: 2D6 + Tactics(naval) + INT DM — opposed check. // B3 p.54

The Captain (or lead tactician) of each ship makes an opposed Tactics(naval) check. The ship with the highest total acts first. In case of a tie, re-roll. Initiative order is fixed for the entire engagement.

How to Roll

Right-click background → **Roll Initiative**. The modal shows all ships.

Ship	Behaviour
Player ships	Enter your 2D6 physical dice in the two fields. Select Tactics skill level and INT DM.
NPC ships	Auto-rolled on confirm.
🎲 button	Opt-in auto-roll for player ships.

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6. Manoeuvre

Any time during a ship's turn, its Pilot may attempt to Open (flee) or Close (approach) against one opposing ship — an **opposed Pilot (DEX) check**, adding the ship's TAC Speed as a fixed DM. // 2300AD B3 p.54

Right-click a ship → **Manoeuvre…**

Control	Action
Acting ship picker	Pick which of the two ships is attempting the manoeuvre.
◀ CLOSE (approach) / OPEN (flee) ▶	Pick the direction, relative to the acting ship.
ROLL 🎲	Both ships roll an opposed Pilot check: 2D6 + Pilot skill + DEX DM + TAC Speed each.
APPLY	On success, the band shifts by the check's Effect in the attempted direction. On failure, the Effect is available for the enemy to close/open instead — the GM decides ("if desired") via a checkbox.
GM SET	Override — sets the band directly, no roll. Use for initial placement.

TAC Speed is a **fixed DM**, never spent or depleted. Pilot's action budget (skill level) can allow multiple manoeuvre attempts in the same round, unlike Gunner which is always capped at 1. // 2300AD B3 p.53–54

Combat ends one round after Distant: if a ship pair is still at Distant range at the start of the round after any resolved Open/Close Pilot check moved them there (not a GM override), a **COMBAT ENDED** badge appears on that pair's DISTANCES row and in the Manoeuvre modal — informational only (doesn't force anything), citing the pursuer's failure to close in time. Use **GM SET** to re-engage the pair and clear it. // 2300AD B3 p.54, issue #46

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7. Attack — Firing Solution

Any ship whose turn it is may attack, once per weapon mount per round (Gunnery is hard-capped at 1 action/round regardless of skill, independently for Turret and Bay Gunner — issue #45). Each weapon slot's `mount` field (`Turret` / `Bay` in the profile editor) picks which Gunner role's skill and action budget the Firing Solution draws from. The attack is a **3-step task chain** — each step's positive Effect carries forward as a DM to the next step. // B3 p.56

Note: the Turret/Bay Gunner distinction is a Trav2022 CRB concept (Crew Duties, p.164) with no 2300AD B3 basis — zero canonical 2300AD ships use a bay-mounted weapon slot. `Bay` mount is kept for homebrew ships only; it isn't a sourced 2300AD mechanic. // issue #63

Right-click a ship → **Attack…**

Step 1 — Sensor Operator

Very Difficult (12+) Electronics (sensors) INT

DM source	Value
Target Signature	+Signature rating
Sensor quality	Basic Military +0 / Improved +1 / Advanced +2
Sensor Time-lag	Adjacent +1 / Close +0 / Short −1 / Medium −2 / Long −3 / Very Long −4 / Distant −5
Target evasion	evasionDm (negative — opposed Pilot check, see §8 Evade; applies here too, not just Step 3)
Engineer assist	Routine (8+) Engineer(power) INT — adds Effect as DM

Step 2 — Pilot

Difficult (10+) Pilot DEX

DM source	Value
TAC Speed	+currentTacSpeed (fixed DM, never spent)
Engineer assist	Routine (8+) Engineer(power) INT
Effect from Step 1	Positive Effect carries forward

Step 3 — Gunner

Difficult (10+) Gunner INT — target number 10+

DM source	Value
Fire Control software	+1 per level (Fire Control/1 = +1, /2 = +2, /3 = +3) — purely additive, never a penalty
Targeting System (per weapon slot)	Light TTA +0, TTA −1, UTES +1, Drone Controller +0 — **DM−8 if the weapon mount has no Targeting System assigned at all**, including Point Defence. Separate from and stacks with Fire Control software. // B3 p.62
Defensive Screens (target's)	−Rating (1/2/3), laser weapons only (`weapon.isLaser === true`) — each laser hit depletes the target's active Rating by 1 // B3 p.62
Range Modifiers	Adjacent +2 / Close +0 / Short −6, applied at the weapon's own max Range and every band below it — e.g. a Short-range weapon still takes −6 at Short, not 0 (`weapon.rangeDm`) // B3 p.57
Atmospheric/planetary condition (target's)	Planetary surface w/atmosphere −6 / w/o atmosphere −4 / atmospheric flight −2 — independent of range band, GM toggle in Ship Detail // B3 p.56
Weapon trait Ortillery	+4 vs. a target on a planetary surface (with or without atmosphere, not in flight) // B3 p.59

DM source	Value
Effect from Step 2	Positive Effect carries forward
EW jamming	−1 (Effect 1–4) / −2 (Effect 5–6) / +1 if the jammer badly failed (Effect ≤−5)
Command (Captain)	+commandBonus, only if the Captain's Command from last round targeted `gunner_turret` (see §9)
Weapon trait Accurate	+1
Weapon trait Slow	−2
Evasion	evasionDm (negative, defender reaction — see §8 Evade)
Captain Tactics Assist	Optional inline roll, Difficult (10+) Tactics(naval) INT — Effect adds to this check only, distinct from Command and not persistent across rounds

Hit = total ≥ 10. Effect = total − 10.

Stationary targets (reaction drive off, in orbit): DM+2, damage doubled. // B3 p.56

Damage

Weapon damage is rolled and reduced by the target's Armour. Net damage is applied to Hull Points.

Weapon trait	Effect
Advanced	+1 damage per die
Obsolete	−1 damage per die
AP X	Ignores X points of Armour

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8. Reactions

The defender can declare reactions before each attack roll.

Evade

Declared via the **Manoeuvre…** modal (right-click → Manoeuvre… → EVADE 🎲) — its effect lasts through the whole round, not just the moment it's rolled. **Opposed Pilot (DEX) check** against the enemy's own Pilot check (Step 2 of the Firing Solution). // B3 p.54

Evading pilot's Effect	Effect on all enemy Electronics(sensors) and Gunner checks
Effect 1–4	DM−1
Effect 5+	DM−2
Effect ≤ −5	Enemy gains DM+1

This is **not** "remaining TAC Speed × Pilot skill" as in the Traveller CRB — it's an active opposed check. `AttackModal` reads the result automatically from `target.evasionDm` in both Step 1 and Step 3; the GM can still override manually if needed.

Point Defence

Declared in the **Drone Attack** modal, against **one incoming drone/missile at a time** — not an abstract salvo. **Difficult (10+) Gunner (DEX)** check.

Weapon type	DM
Conventional mount	DM−2
PDC (e.g. Quinn Type 17)	DM+4

Success destroys that specific drone before it can attack.

Gunner-managed vs Computer-run (issue #57): a PDC-equipped intercepting weapon shows a mode toggle. Gunner-managed (default) is the DM table above, capped at 1/round like every other Gunnery action. Computer-run drops the crew skill/DEX entirely — just Fire Control software + Targeting System hardware + a flat DM+4 — and can be used up to **TL−4** times per round (the ship's own Tech Level, now a real tracked field), independent of the shared Gunnery budget. Both share the same Difficult (10+) target and the same one-drone-at-a-time resolution.

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9. Crew Actions

Any time during a ship's turn, each crew role may spend its own remaining actions on a crew action — right-click a ship → **Crew Action…** The Captain can also spend a Leadership action to grant another role +1 action this round (**Issue Order**), distinct from Commands below.

Select the action type, choose applicable target/options, roll (if required), then click **APPLY RESULT**.

Captain

Action	Check	Effect
Commands	Routine (8+) Leadership INT/SOC	Order one crew role. Effect 1–4 → DM+1, Effect 5–6 → DM+2 to their actions. Active **immediately, this round** — the Captain acts first among the crew, so a Command issued early in the ship's turn is already available to that role's actions later the same round. Resets at the start of the next round if not re-issued. Auto-applied if targeting `gunner_turret` (Step 3 Gunner check) or `pilot` (Evade roll); for other roles the GM adds it manually. **Crew Discipline** (issue #39): mark a role as having disobeyed the order instead of receiving it — DM−1 to their actions this round, no check, doesn't consume the Captain's Leadership cap. // 2300AD B3 p.54
Tactics Assist	Difficult (10+) Tactics(naval)	Optional inline roll inside the Attack modal's Step 3 — adds its Effect to that single Gunner check

Action	Check	Effect
	INT	only. Distinct from Commands, stacks with it.

Engineer

Action	Check	Effect
Overload Stutterwarp	Very Difficult (10+) Engineer(stutterwarp) INT — B3 prints this label at a 10+ threshold, an inconsistency with the book's other "Very Difficult" checks (always 12+ elsewhere), likely an erratum; threshold unchanged, issue #77	Success: Effect 1–4 → TAC Speed +1, Effect 5–6 → +2 this round. No failure consequence.
Boost Power Output	Difficult (10+) Engineer(power) EDU	Repeated every round. Success: Effect is the % Power increase — informational only, no Power resource tracked. Effect ≤−5: automatic critical hit to the Power Plant. // 2300AD B3 p.54
Emergency Repair	Difficult (10+) Mechanic INT/EDU	*System mode*: reduce one critical hit severity by 1 — **temporary**: Effect < 5 holds for 1D rounds, Effect 5+ holds for the rest of combat, then relapses back up by 1 (issue #64). *Hull mode*: restore 5 Hull Points — permanent.

Sensor Operator

Action	Check	Effect
Electronic Warfare	Difficult (10+) Electronics(comms) EDU	Effect 1–4: jammed ship suffers DM−1 to Gunner checks. Effect 5–6: DM−2. Effect ≤−5: jam backfires, jammed ship instead gains DM+1. Resets at round end. Target ship's Encrypted Comms toggle applies DM−2 to this roll (B3 p.58, issue #66). // 2300AD B3 p.54, p.58, issue #73
Active Sensors	Easy (6+) Electronics(sensors)	Activates active sensor sweep. Signature +1 while active (flag persists until toggled off in Ship Sheet).

Gunner

Point Defence is a reaction inside the Drone Attack modal — see §8. It is not a Crew Action menu entry, since it must be declared against one specific incoming drone, not chosen freely. Gunner has no other Crew Action beyond Fire Weapon / Deploy-Recharge Screens / Point Defence / Operate UTES Array, all sharing the same single Gunnery action per round.

Evasion (opposed Pilot check, B3 p.54) is resolved directly in the **Manoeuvre…** modal (see §6/§8), not as a Crew Action — Pilot has no entry in this menu.

Marines / Ship's Troops

Action	Check	Effect
Boarding Action	No skill check — flat 2D6 + modifiers	Two-roll opposed resolution (see §12 Boarding).
Repel Boarders	No skill check — flat 2D6 + modifiers	Defender's roll for boarding resolution (see §12 Boarding).

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10. Critical Hits

Surface Fixture Damage // B3 p.58

Triggered on any hit with **Effect ≥ 3**, even if it does not penetrate armour.

Roll 2D on the Surface Fixture table:

2D	System	1st Hit	2nd Hit
2	Fire Control	DM−2 to attack rolls	—
3–4	Weapon	−1D damage, DM−2 to attacks	Disabled
5	Sensors	DM−2 to Electronics(sensors)	—
6–8	Radiator	See Radiator rules	—
9	Sensors	DM−2 to Electronics(sensors)	—
10–11	Discharge Vanes	Disabled	Destroyed
12	Other System	Disabled	Destroyed

3rd hit: B3 p.58 prose (separate from the table above): "All systems except radiators are destroyed on the third hit. Checks that require a system that has been destroyed are not possible." Applies to Fire Control/Weapon/Sensors — Discharge Vanes/Other System already reach Destroyed at the 2nd hit; Radiator has its own 5-stage escalation instead.

Internal Critical Hits // B3 p.58 (cites CRB "p.158–159" — actually p.169/p.170 in this PDF, issue #23/#67)

Triggered when the attack Effect is ≥ 6 (B3-sourced), or when hull reaches 0 (house rule — not B3/CRB-sourced, kept for narration on a killing blow // issue #76).

Roll 2D for location (CRB p.169); per-severity effects come from the Effects table (CRB p.170). 2300AD substitutions: - **J-Drive** → **Stutterwarp Drive** (severity reduces TAC Speed by −1 per severity level) - **M-Drive** → **Reaction Drive** (first crit: inoperable; second: destroyed)

Active Hazards

Some critical hits generate ongoing hazards (fire, hull breach, fuel leak, radiation exposure). Purely GM-managed bookkeeping — add/remove them directly via the **Ship Sheet → ACTIVE HAZARDS** panel, no skill check gates this.

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11. Drones & Missiles

2300AD B3 has no "salvo" abstraction — each drone/missile is an **individually piloted unit** that closes range on its own TAC Speed and resolves its own 3-step Firing Solution, exactly like a ship. // B3 p.55–56, p.61

Launching

Right-click attacker → **Launch Drone…** Pick a target, a weapon (Ritage-1, Ritage-2, 'Whiskey', or a vehicle-sourced missile), and how many separate units to launch.

Each unit is tracked independently in the **Drone Tracker** with its own TAC Speed and Endurance (maximum rounds before it goes inert if it never reaches its target). Each round it closes one range band toward its target.

No canonical combat drone carries the Smart trait (B3 p.61 lists none for any of the three — only the vehicle-sourced missiles `aero12` / `kingfisher` have it, issue #50/#51).

Drone	TL	Magazine	Damage	TAC Speed	Endurance	Traits
Ritage-1	11	5	1D	3	60 rounds (6h)	—
Ritage-2	12	— (single-shot)	5D	4	40 rounds (4h)	Blast 6, Radiation
'Whiskey'	12	3 (battery) / single-use (detonation)	1D laser / 3D detonation	4	20 rounds (2h)	— / Blast 3, Radiation

Resolving an Attack

Once a ship's own drone reaches Close/Adjacent range, right-click that ship → **Resolve drone attack…** (or click the drone in the Drone Tracker). This opens the same 3-step Firing Solution as a normal ship attack:

Step	Who	Check	DM
1 — Sensor	Sensor hand-off (no penalty, no lag) or Remote Pilot self-generated (Piloting action, DM−2, plus the drone's own flat DM+2 Pilot bonus and lightspeed lag)	Very Difficult (12+)	+Signature, sensor quality, time-lag
2 — Position Vessel	Remote Pilot	Difficult (10+), Electronics(remote ops) DEX	+drone TAC Speed, +carry from Step 1, +lightspeed lag
3 — Gunner	—	Difficult (10+)	+Fire Control, +range DM at drone's current band, +carry from Step 2, +target's reactive DMs, +lightspeed lag

Lightspeed lag (B3 p.55, issue #49): DM−1 to a drone's own actions once it's Long range or farther from its *controlling ship* (not its target — that's the Step 1 time-lag above). A hand-off Sensor check is exempt (per B3's own text); self-generated Sensor, Position Vessel, and Gunner are not.

Point Defence

Resolved inline, at the top of the same modal, before the Firing Solution — one drone at a time. See §8. Success destroys that drone before it can attack.

Impact

Damage is rolled the same way as a normal weapon hit (Effect does **not** add to damage — B3 p.56 note). After the attack resolves (hit or miss), the drone consumes one shot from its magazine (B3 p.61 Combat Drones table): Ritage-1 (5 shots) and Whiskey's battery-laser mode (3 shots) survive to attack again in a later round until the magazine is exhausted; Ritage-2 and Whiskey's detonation mode are always single-use nuclear warheads that destroy the drone outright, regardless of remaining battery charge.

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12. Boarding

Boarding resolves in **one opposed roll per round**, continuing until one side wins or retreats. Only at **Adjacent range**, spent from the Marines' own action budget.

Attacker's Roll (Boarding Action)

2D6 + modifiers. No skill check. // Trav2022 CRB p.175

Defender's Roll (Repel Boarders)

2D6 + modifiers. No skill check. // Trav2022 CRB p.175

Modifiers

Condition	DM
Superior armour	+1
Superior weapons	+1
Superior skill/tactics	+2
Outnumber defender	+1
Vastly outnumber defender	+3
Defender without assigned marines	−2

Result Table (Attacker − Defender)

Difference	Result	Hull damage
−7 or less	Attackers defeated; defender may counter-attack DM+4	—
−4 to −6	Attackers defeated; must retreat or are captured	—
−1 to −3	Combat continues; defender DM+2 next round	2D Hull to defending ship
0	Combat continues; no advantage	—
1 to 3	Combat continues; attacker DM+2 next round	2D Hull to defending ship
4 to 6	Boarding succeeded; 2D rounds to pacify	1D Hull (ignores Armour)
7 or more	Immediate boarding; ship control passes to attackers	—

The GM applies the DM carry-over manually as a modifier in the next round's boarding roll.

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13. Signature

Every ship has a base **Signature** value. Used as a positive DM in enemy Electronics(sensors) checks during the Firing Solution (Step 1). // B3 p.57

The effective Signature is computed dynamically — open the **Ship Sheet** to see the breakdown and toggle conditions.

Automatic Modifiers

Condition	Signature DM
Hull damage > 50%	+1
Power Plant critical hit	+1
EW active on this ship	+2

GM Toggles (Ship Sheet → Signature Conditions)

Condition	DM
Radiators Retracted	−1
Heat Sink Active	−4
Solar Panels Extended	+2
Spin Habitat Retracted	−1
Reaction Drive Active	+4 / +6 / +8 (rockets / thrusters / nuclear)
Active Sensors On	+1
Stealth system	−4

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14. Undo / Redo

Every game state change pushes a snapshot to the undo stack (50-step maximum).

Control	Action
↩ Undo	Restore the previous state.
↪ Redo	Re-apply an undone action.

Both buttons appear in the HUD only when their stacks are non-empty. The battle log is not rolled back — an Undo entry is appended instead.

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15. Save & Resume

Autosave

The app autosaves to IndexedDB after every significant action. On return, click 🔄 **RESUME** on the Dashboard to restore instantly.

Manual Save

Click 💾 **SAVE** in the HUD to download the full session as a `.json` file.

Resume From File

On the Dashboard, click ↓ **RESUME FROM FILE** and select a `.json` file. A preview shows the full roster before you confirm loading.

Profile Export / Import

Ship profiles are independent from battle sessions. Use ↑ **EXPORT** and ↓ **IMPORT** in the profile panel to share or back up profiles separately.

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